

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	26 May 2026
Team ID	HDI Predictor Team
Project Name	HDI Predictor using Machine Learning and Flask
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Preparation	Load HDI.csv, select prediction columns, and save the cleaned dataset.
FR-2	Prediction Input Handling	Accept country and indicator values through the Flask form and API.
FR-3	Model Inference	Load HDI.pkl and produce the HDI score, category, and confidence.
FR-4	Visualization	Render charts and progress indicators on the result page.
FR-5	Batch Prediction	Process CSV uploads and return HDI predictions for multiple rows.
FR-6	Country Auto-Fill	Return indicator values for a selected country from the cleaned dataset.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The browser interface should be simple, responsive, and easy to understand.
NFR-2	Performance	Single predictions should respond quickly and batch predictions should complete without delay.
NFR-3	Reliability	The model should load consistently and return stable outputs for valid inputs.
NFR-4	Maintainability	Notebook, model, routes, templates, and static files should stay modular.
NFR-5	Security	Input and upload validation should prevent malformed requests and file errors.
NFR-6	Scalability	The solution should allow future datasets, features, and model improvements.